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Course: Calculus Beta

Assignment: MyLab Math - Uge 12

Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

$$f(x,y) = \sqrt{6x^2 + 7y^2}$$

The symbol $\frac{\partial f}{\partial x}$ means the partial derivative of f with respect to x . To find this derivative, treat y as a constant and differentiate the given function f with respect to x in the usual way.

Use the Chain Rule to differentiate the function. To begin, differentiate f by treating it as the square root of a function of x .

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} \sqrt{6x^2 + 7y^2} \\ &= \frac{1}{2\sqrt{6x^2 + 7y^2}} \frac{\partial}{\partial x} (6x^2 + 7y^2)\end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned}\frac{\partial f}{\partial x} &= \frac{1}{2\sqrt{6x^2 + 7y^2}} \frac{\partial}{\partial x} (6x^2 + 7y^2) \\ &= \frac{1}{2\sqrt{6x^2 + 7y^2}} (12x) && \text{Evaluate.} \\ &= \frac{6x}{\sqrt{6x^2 + 7y^2}} && \text{Simplify.}\end{aligned}$$

The symbol $\frac{\partial f}{\partial y}$ means the partial derivative of f with respect to y . To find this derivative, treat x as a constant and differentiate the given function f with respect to y in the usual way.

Use the Chain Rule to differentiate the function. To begin, differentiate f by treating it as the square root of a function of y .

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} \sqrt{6x^2 + 7y^2} \\ &= \frac{1}{2\sqrt{6x^2 + 7y^2}} \frac{\partial}{\partial y} (6x^2 + 7y^2)\end{aligned}$$

Then evaluate the partial derivative in the result.

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{1}{2\sqrt{6x^2 + 7y^2}} \frac{\partial}{\partial y} (6x^2 + 7y^2) \\ &= \frac{1}{2\sqrt{6x^2 + 7y^2}} (14y) && \text{Evaluate.} \\ &= \frac{7y}{\sqrt{6x^2 + 7y^2}} && \text{Simplify.}\end{aligned}$$

Thus, the partial derivatives are shown below.

$$\frac{\partial f}{\partial x} = \frac{6x}{\sqrt{6x^2 + 7y^2}} \qquad \frac{\partial f}{\partial y} = \frac{7y}{\sqrt{6x^2 + 7y^2}}$$